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Routing: Link State and Dijkstra’s Algorithm and Routing: Distance Vector, Bellman-Ford, and BGP Introduction

This report provides a detailed examination of the core routing methodologies discussed in Lectures 15 and 16, covering the transition from intra-domain link-state and distance-vector algorithms to the inter-domain complexities of the Border Gateway Protocol (BGP)

1. Link-State Routing and Dijkstra’s Algorithm
The primary objective of link-state routing is for **e**very router to obtain a complete, identical map of the network topology. This family of algorithms is characterized by the distribution of local information globally .

The Graph Abstraction: Routers are modeled as vertices (*V*), and physical or logical connections are modeled as edges (*E*). Edge weights (costs) are assigned based on various metrics: hop count (RIP), inverse bandwidth (OSPF), or administrative preference. For example, the OSPF default cost is 10^8/B which directly relates cost to transmission delay.
The Three-Phase Process:

1. Discovery: Routers use "hello packets" to identify neighbors and link costs.
2. Flooding: Routers generate Link-State Advertisements (LSAs) and flood them throughout the network. Reliable flooding is ensured through sequence numbers to prevent infinite loops and age timers to refresh stale data.
3. Shortest-Path Computation: Once a router has the full Link-State Database (LSDB), it independently runs Dijkstra’s algorithm to produce a Shortest-Path Tree (SPT) rooted at itself

OSPF (Open Shortest Path First): This is the most widely deployed link-state protocol. To scale to large networks, OSPF uses **Areas** (specifically Area 0 for the backbone) to contain LSA flooding and reduce the size of the LSDB.

The Oscillation Problem: A significant risk in link-state routing occurs if costs are load-sensitive. Traffic can shift back and forth between paths as they become alternately congested and idle, a cycle that never stabilizes

2. Distance-Vector Routing and Bellman-Ford
Distance-vector (DV) routing represents a "dual" approach to link-state: instead of global topology knowledge, **routers distribute global distance information locally** to their immediate neighbors only

The Bellman-Ford Equation: The mathematical foundation of DV routing is dx(y) = min vneighbors(*x*) {c(x,v)+dv(y)} This implies the shortest path from *x* to *y* is found by choosing the neighbor *v* that offers the minimum sum of the direct link cost and that neighbor's own reported distance to the destination

Asymmetric Convergence: A defining characteristic of DV is that "good news travels fast, but bad news travels slow". When a link cost decreases, improvements propagate quickly. However, link failures can lead to the Count-to-Infinity problem, where routers update each other in a loop, incrementing costs indefinitely because they lack a global view to detect circular dependencies Mitigation Techniques: To combat routing loops, protocols use Split Horizon (not advertising a route back to the neighbor it was learned from) and Poisoned Reverse (actively advertising a cost of ∞ to that neighbor).

RIP (Routing Information Protocol): The simplest DV protocol, RIP uses a **hop count metric** and defines ∞ as 16, limiting its use to networks with fewer than 15 hops.

3. Scaling to the Global Internet: Autonomous Systems and BGP

Neither OSPF nor RIP can scale to the entire internet due to the massive size of the global topology and the need for administrative autonomy. The solution is hierarchical routing.

Autonomous Systems (AS): The internet is a collection of over 75,000 independent networks called Autonomous Systems, each controlled by a single organization (e.g.Comcast, Google).

The Border Gateway Protocol (BGP): BGP is the "glue" that connects these ASes. Unlike interior protocols that seek the shortest path, BGP is a Path Vector protocol driven by business relationships and policy.

AS-PATH and Loop Prevention: BGP advertisements include an AS-PATH attribute listing every AS the route has crossed. If a router sees its own AS Number (ASN) in the path, it rejects the route, preventing loops without the need for a count-to-infinity mechanism.

Business Policies: Routing decisions are often based on economics:

- **Customer–Provider:** The customer pays for transit.
- **Peer–Peer:** Two ASes exchange traffic for free, but only for their respective customers.
- **Valley-Free Rule:** A well-configured BGP system ensures traffic only flows from customer to provider or between peers, never providing free transit to a peer's provider.

Security and Stability: Because BGP relies on trust, it is vulnerable to BGP Hijacking, such as the 2008 Pakistan/YouTube incident, where a more-specific prefix announcement redirected global traffic to a black hole

Summary of the Routing Landscape

The current internet architecture utilizes a tiered approach: Intra-AS routing (OSPF/RIP) focuses on finding the shortest path within an organization, while Inter-AS routing (BGP) manages reachability and policy between organizations. While link-state protocols like OSPF have largely replaced distance-vector protocols like RIP in production for their faster convergence and richer metrics, both remain fundamental to understanding the control plane

Personal reflectin:
The "Pakistan/YouTube" incident of 2008 is perhaps the most eye-opening example of the human and political element in networking. Seeing the Algorithms like Dijkstra and Bellman that operate on perfect logic it was fascinating to learn that the global internet effectively runs on an agreement between engineers in the 1980s. It is wild to realize that a simple change from a /22 to a /24prefix can effectively hijack global traffic. This demonstrates how the Longest Prefix Match rule, while efficient for routing, is a massive security loophole if the person announcing the prefix is lying. It is really fascinating to see just how unstructured the global internet is, I thought it was strict structured with many rules but it’s quit the opposite it’s just routers asking each other hoping the answer’s is true and that in the world of networking business policy and trust matter more than math at times (I guess that’s also many other things).